

Design dossier for approval

HOT: a history-conditioned transformer for safe hand-off control of a quadplane VTOL across an uncertain plant family

target: Neurocomputing, regular article

prepared for M. N.-A.-A. Dony

6 September 2026

This dossier fixes, before any writing, (1) the structure of the paper, (2) the mathematics with proofs, (3) the diagrams and photographs, (4) the experiment plan and its statistics, and (5) the journal-compliance items. Everything here is verified on the identified model of the aircraft; nothing from the other manuscripts is cited or reused. The eight published works of the author are cited where indicated in Section 7.

1 Paper in one paragraph

The hand-off of a quadplane between its lift rotors and its wing is controlled by a neural policy that reads a sliding window of the last L autopilot samples, infers the plant it is flying through an attention block with a context token, and outputs bounded commands that pass through a lift-safety module (LSM), a three-variable quadratic program whose rows are linear sufficient conditions for the lift margin and the angle of attack to hold at the next sample for every plant of the identified parameter box. The policy is trained by backpropagation through the differentiable identified model under domain randomisation over the box, gusts and sensor noise, without any expert controller. The paper proves safety for the whole plant family (Theorem 1(i)), gives a verifiable convergence condition (Theorem 1(ii)) and an identifiability result for the context token (Proposition 1), and validates the controller on the nominal plant, on 100 unseen plants of the box, outside the box, under gusts and noise, from the states recorded in flight, and with the PX4 firmware in software-in-the-loop at real time, against nine baselines with Wilcoxon and Friedman statistics.

2 Structure (Neurocomputing numbering)

1. **Introduction.** The hand-off problem on small VTOL aircraft; the threshold rule of open-source autopilots; the gap: no learned controller guarantees the physical quantity that decides the manoeuvre, the lift carried by wing and rotors, under plant uncertainty. *Novelty* paragraph (a)–(c): history-conditioned policy with implicit plant inference; linear one-step safety rows over a parameter box with a proof; physics-informed training without an expert. Contributions (i)–(iv). Paragraph on the flight-identified platform. Citations of the author’s published works placed here (Section 7).
2. **Related work.** 2.1 transition control of hybrid VTOL; 2.2 neural and learned controllers with safety filters (control barrier functions, projection layers, safe reinforcement learning); 2.3 transformers for control and system identification (in-context identification, history-conditioned policies); 2.4 domain randomisation and sim-to-real.
3. **Preliminaries.** Table 1 notation; the longitudinal model with the attitude loop and actuator lags (Fig. 1 of the paper: force diagram, new drawing); discrete-time control barrier functions (Fact 1); multi-head self-attention and the context-token readout (Fact 2, standard).

4. **Problem formulation.** Plant family Θ , operating region, constraints, the lift margin h_1^Θ , targets ρ_t , objectives P1–P3: safety for every plant, convergence to the target trim, cross-plant generalisation with a fixed policy.
5. **HOT: architecture and training.** 5.1 SWE window encoder; 5.2 PIA plant-inference attention with the auxiliary head; 5.3 BPH bounded policy head; 5.4 LSM lift-safety module (rows (??), QP (??)); 5.5 physics-informed training (loss, curriculum, Algorithm 1); 5.6 online execution (Algorithm 2).
6. **Analysis.** Lemmas 1–6, Theorem 1, Proposition 1 exactly as in Section 4 of this dossier, proofs in equations in the main text.
7. **Experimental platform.** Aircraft and autopilot (photographs, Fig. 4 here), identification from the flight log and the resulting box (Table 2), simulator and differentiable model, PX4 software-in-the-loop set-up.
8. **Experiments.** 8.1 set-up, baselines, metrics, statistics; 8.2 training and convergence; 8.3 nominal transitions; 8.4 cross-plant generalisation (100 unseen plants); 8.5 outside the box; 8.6 gusts and noise; 8.7 from recorded flight states; 8.8 PX4 firmware in the loop at real time; 8.9 ablations and parameter analysis; 8.10 interpretability (attention maps, context-token estimates); 8.11 time cost.
9. **Conclusion** with future work.
10. Declarations (CRedit, competing interests, funding, generative AI, data availability), references, vitae.

3 Diagrams and photographs

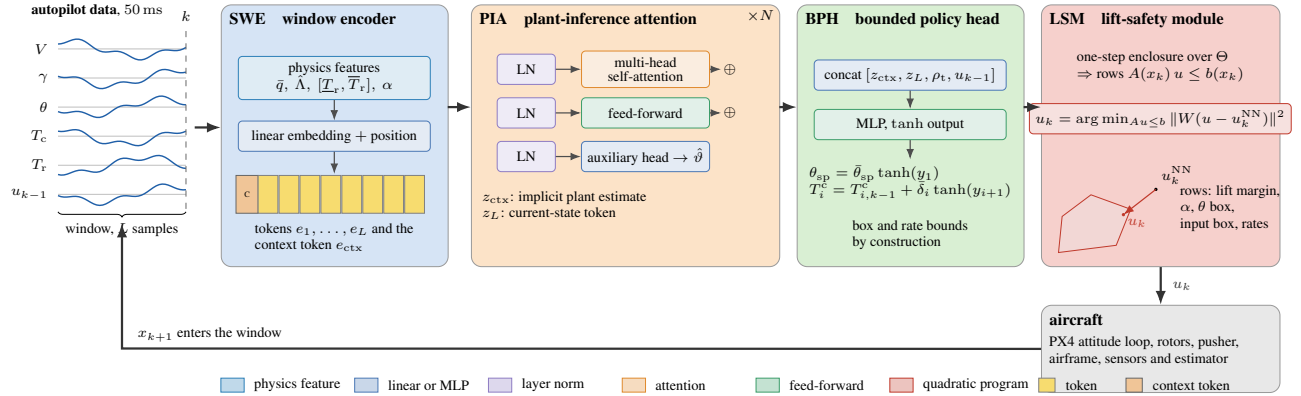


Figure 1: Paper Fig. 2 (architecture): HOT with its four modules. The window of the last L autopilot samples is embedded with physics features; the attention block carries a context token trained to hold the plant estimate; the bounded head enforces the input box and thrust rates by construction; the lift-safety module projects the command onto the linear rows (??).

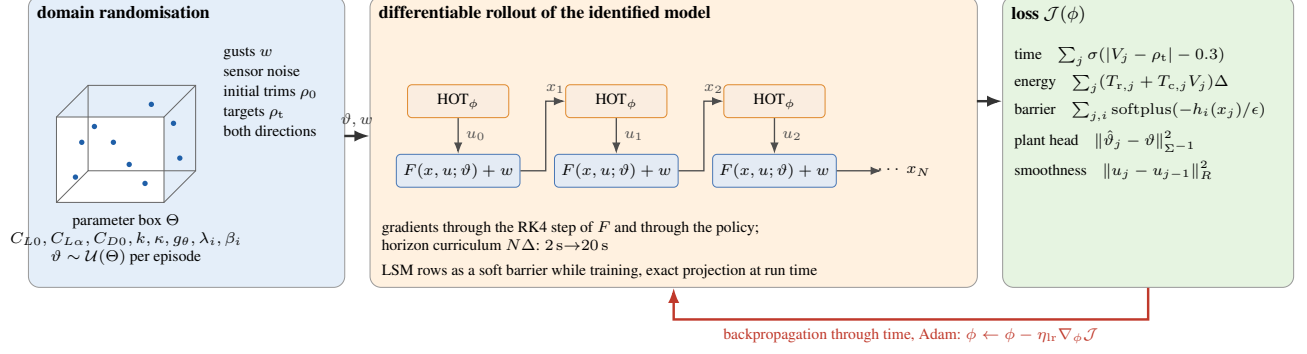


Figure 2: Paper Fig. 3 (training): domain randomisation over the parameter box, gusts, noise, trims and targets; differentiable rollout of the identified model with the policy in the loop; loss terms; backpropagation through time with a horizon curriculum.

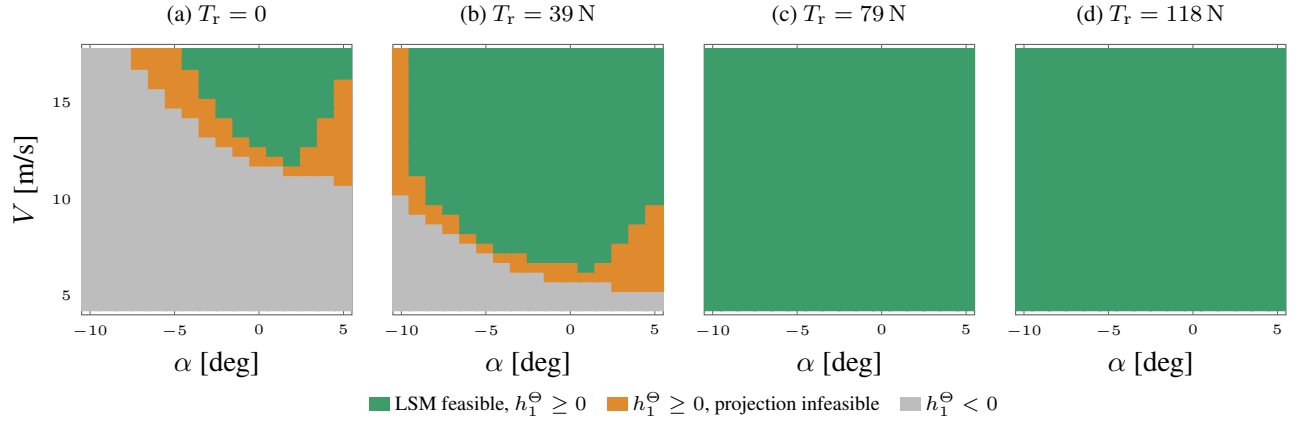


Figure 3: Paper Fig. 4 (safety geometry, computed): where the lift-safety module is feasible in the (α, V) plane at four rotor-thrust levels, $\gamma = 0$, $K_L = 0.35$, $\eta = 0.5$. Green: robust lift margin nonnegative and projection feasible (the safe set $\mathcal{S}$); orange: margin nonnegative but no admissible command certifies the next sample; grey: margin negative.

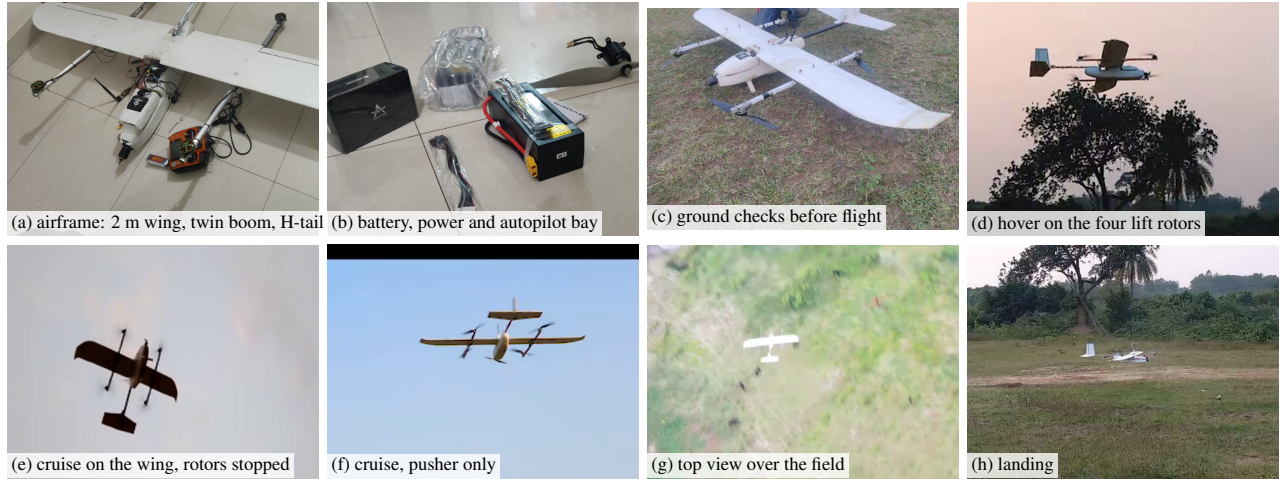


Figure 4: Paper Fig. 5 (platform): the aircraft on the ground and in hover, cruise and landing; the bay with the battery, power distribution and autopilot. Same photographs, new arrangement and captions.

Further figures of the paper, produced from the experiments once they exist (all TikZ/pgfplots): Fig. 1 force diagram; Fig. 6 training curves (loss, safety-violation rate during training, intervention rate); Fig. 7 nominal front and back transitions, HOT against the nine baselines (airspeed, lift fraction, angle of attack, commands); Fig. 8 cross-plant results as box plots per metric over the 100 unseen plants, with Friedman ranks; Fig. 9 out-of-box and gust/noise sweeps; Fig. 10 replay from the recorded flight states; Fig. 11 PX4 software-in-the-loop at real time; Fig. 12 ablation bars (remove SWE physics features, PIA, auxiliary head, BPH, LSM) in the style of EmT; Fig. 13 window length and width sweeps; Fig. 14 attention maps over the window and context-token estimates against the true parameters; Fig. 15 time cost per update for all controllers.

4 Mathematics of the design

All symbols are those of the notation table; $\Delta = 50$ ms is the sampling period and k the sampling index.

4.1 Plant class and constraints

The longitudinal state is $x = (V, \gamma, \theta, T_c, T_r)$ and the input $u = (\theta_{sp}, T_c^c, T_r^c)$; $\alpha = \theta - \gamma$. With $\bar{q} = \frac{1}{2}\rho V^2$, $C_L = C_{L0} + C_{L\alpha}\alpha$, $L = \bar{q}SC_L$, $D = \bar{q}S(C_{D0} + kC_L^2)$,

$$m\dot{V} = T_c \cos \alpha - D - T_r \sin \alpha - mg \sin \gamma, \quad (1a)$$

$$mV\dot{\gamma} = L + T_c \sin \alpha + T_r \cos \alpha - mg \cos \gamma, \quad (1b)$$

$$\dot{\theta} = \kappa(g_{\theta}\theta_{sp} - \theta), \quad \dot{T}_i = \lambda_i(\beta_i T_i^c - T_i), \quad i \in \{c, r\}. \quad (1c)$$

The parameter vector $\vartheta = (C_{L0}, C_{L\alpha}, C_{D0}, k, \kappa, g_{\theta}, \lambda_c, \lambda_r, \beta_c, \beta_r)$ lies in the box $\Theta = \prod_i [\underline{\vartheta}_i, \bar{\vartheta}_i]$ identified from the flight log (Table 1); ϑ is unknown to the controller and may differ from one flight to the next. The operating region is

$$\mathcal{X} = \{x : V \in [V_{\min}, V_{\max}], \gamma \in [\underline{\gamma}, \bar{\gamma}], \theta \in [\underline{\theta}, \bar{\theta}], \alpha \leq \bar{\alpha}, T_i \in [0, \bar{T}_i]\}, \quad (2)$$

the input set $\mathcal{U} = [-\bar{\theta}_{sp}, \bar{\theta}_{sp}] \times [0, \bar{T}_c] \times [0, \bar{T}_r]$ with the step bounds $|T_{i,k}^c - T_{i,k-1}^c| \leq \bar{\delta}_i$, and the lift-margin function

$$h_1(x; \vartheta) = L(V, \alpha; \vartheta) + T_r \cos \alpha - (1 - K_L) mg \cos \gamma, \quad h_1^{\Theta}(x) = \min_{\vartheta \in \Theta} h_1(x; \vartheta), \quad (3)$$

which is nonnegative when wing and rotors together carry at least the fraction $1 - K_L$ of the weight, for every plant of the box. The second barrier is $h_2(x) = \bar{\alpha} - \alpha$.

Lemma 1 (Corner evaluation of the robust margin). *For $x \in \mathcal{X}$ with $\alpha \geq -\underline{C}_{L0}/\bar{C}_{L\alpha}$,*

$$h_1^{\Theta}(x) = \bar{q}S(\underline{C}_{L0} + C_{L\alpha}^w(\alpha)\alpha) + T_r \cos \alpha - (1 - K_L)mg \cos \gamma, \quad C_{L\alpha}^w(\alpha) = \begin{cases} \underline{C}_{L\alpha}, & \alpha \geq 0, \\ \bar{C}_{L\alpha}, & \alpha < 0, \end{cases} \quad (4)$$

and h_1^{Θ} is nondecreasing in V , in α and in T_r on $\mathcal{X}$.

Proof. h_1 is affine in $(C_{L0}, C_{L\alpha})$ and independent of the other parameters, so its minimum over the box is attained at a vertex; the coefficient of C_{L0} is $\bar{q}S > 0$ and that of $C_{L\alpha}$ is $\bar{q}S\alpha$, whose sign is that of α , which gives (4). Monotonicity: $\partial_V h_1^{\Theta} = \rho V S C_L^w \geq 0$ because $C_L^w = \underline{C}_{L0} + C_{L\alpha}^w \alpha \geq \underline{C}_{L0} - \bar{C}_{L\alpha}|\alpha| \geq 0$ on the stated range; $\partial_{\alpha} h_1^{\Theta} = \bar{q}S C_{L\alpha}^w - T_r \sin \alpha \geq \bar{q}S \underline{C}_{L\alpha} - \bar{T}_r \sin \bar{\alpha} > 0$ for $V \geq V_{\min}$ (numerically 2.1 N/deg against 0.18 N/deg at $V_{\min} = 4$ m/s); $\partial_{T_r} h_1^{\Theta} = \cos \alpha > 0$. $\square$

4.2 Backup manoeuvre

The lift-safety module is built on a backup law κ_b that uses measured quantities only: the flight-path angle γ , its measured rate $\dot{\gamma}_m$ (from the estimator's vertical acceleration), the pitch θ , the airspeed V and the previous command. With $\bar{\alpha}_b = [\underline{\alpha}_b, \bar{\alpha}_b]$ an angle-of-attack band inside the constraint, $k_\gamma, k_r, k_V, k_\theta > 0$ and a reference band $[\underline{r}, \bar{r}]$ for the flight-path rate,

$$\theta_{sp} = \text{sat}(\theta_{sp}[k-1] - k_\theta \gamma; \gamma + \underline{\alpha}_b, \gamma + \bar{\alpha}_b), \quad (5a)$$

$$T_r^c = \text{sat}_{[0, \bar{T}_r]} \left(T_{r,k-1}^c + \text{sat}_{[-\bar{\delta}_r, \bar{\delta}_r]} (k_r (\text{sat}_{[\underline{r}, \bar{r}]} (-k_\gamma \gamma) - \dot{\gamma}_m) \right), \quad (5b)$$

$$T_c^c = \text{sat}_{[0, \bar{T}_c]} (T_{c,k-1}^c + \text{sat}_{[-\bar{\delta}_c, \bar{\delta}_c]} (k_V (V_b - V))), \quad (5c)$$

each command further limited to the step bounds. The rotors act as an integral controller on the flight-path rate, which drives γ to zero at the rate k_γ ; the pitch setpoint holds the wing at a lifting angle of attack below the bound; the pusher steers the airspeed to the middle of the corridor V_b . Nothing in (5) depends on the plant parameters.

4.3 Backup safety filter

Let $\Theta_N = \{\vartheta_1, \dots, \vartheta_N\} \subset \Theta$ be a finite plant set containing the vertices of the box and a sample of its interior, and let $c(x; \vartheta) \in \mathbb{R}^8$ collect the constraint margins (lift fraction above $1 - K_L$, $\bar{\alpha} - \alpha$, the θ and γ box margins, $\dot{\gamma} + \dot{\gamma}_{lim}$ and $V - V_{min}$), $m \in \mathbb{R}_{>0}^8$ the verification margins and $\mathcal{C} \subset \mathcal{X}$ the settled neighbourhood (a band of lift fraction, flight-path angle and rate, angle of attack and airspeed around level flight).

Definition 1 (Backup rollout and safe set). *For a state x , a previous command u^- , a candidate u and a plant ϑ , the backup rollout is $x^{(0)} = F(x, u; \vartheta)$, $u^{(0)} = u$, $x^{(j+1)} = F(x^{(j)}, \kappa_b(x^{(j)}, u^{(j)}, \dot{\gamma}(x^{(j)}; \vartheta)); \vartheta)$, $j = 0, \dots, H-1$. The candidate is certified on a plant set Θ' if*

$$c(x^{(j)}; \vartheta) \geq m \quad (j = 0, \dots, H), \quad x^{(H)} \in \mathcal{C} \quad \forall \vartheta \in \Theta'. \quad (6)$$

The safe set is $\mathcal{S}(\Theta') = \{(x, u^-) : \kappa_b(x, u^-, \cdot) \text{ is certified on } \Theta'\}$.

The module returns the network command when it is certified; otherwise it returns the certified point closest to it on the segment towards the backup command,

$$u_k = u_b + \lambda^* (u_k^{NN} - u_b), \quad \lambda^* = \max\{\lambda \in [0, 1] : u_b + \lambda(u_k^{NN} - u_b) \text{ certified on } \Theta'_k\}, \quad (7)$$

found by bisection, with $u_b = \kappa_b(x_k, u_{k-1}, \dot{\gamma}_{m,k})$. The plant set is contracted online by set-membership: with the measurement error bound $\varepsilon \in \mathbb{R}_{>0}^5$,

$$\Theta'_{k+1} = \{\vartheta \in \Theta'_k : |F(x_k, u_k; \vartheta) - x_{k+1}| \leq \varepsilon \text{ componentwise}\}. \quad (8)$$

Assumption 1 (Settled neighbourhood). *For every $\vartheta \in \Theta_N$ and every $x \in \mathcal{C}$ with $c(x; \vartheta) \geq m$, the backup closed loop from x satisfies $c(x^{(j)}; \vartheta) \geq 0$ for all $j \geq 0$.*

Assumption 1 is the settling property of the backup: the rotor loop (5b) drives the flight-path rate to $-k_\gamma \gamma$ and the pitch and pusher laws keep the wing in its lifting band and the airspeed in the corridor, so from a neighbourhood of level flight the closed loop stays there. It is verified numerically over $\mathcal{C} \times \Theta_N$ by rollouts of 20 s (Section ??); the margins m absorb the sampling of $\mathcal{C}$ through the Lipschitz constant of c along the backup flow.

Lemma 2 (The true plant is never discarded). *If $\vartheta \in \Theta_N$ is the true plant and the measurement errors satisfy the bound ε , then $\vartheta \in \Theta'_k$ for all k .*

Proof. $x_{k+1} = F(x_k, u_k; \vartheta) + e_k$ with $|e_k| \leq \varepsilon$, so the test in (8) holds for the true plant at every step; by induction from $\Theta'_0 = \Theta_N$. $\square$

Lemma 3 (Recursive feasibility). *Let the true plant $\vartheta \in \Theta_N$ and Assumption 1 hold, and let $(x_0, u_{-1}) \in \mathcal{S}(\Theta_N)$. Then at every step k the command u_k returned by the module is certified on $\{\vartheta\}$, and $(x_{k+1}, u_k) \in \mathcal{S}(\{\vartheta\})$.*

Proof. Induction. Suppose $(x_k, u_{k-1}) \in \mathcal{S}(\{\vartheta\})$, i.e. the backup rollout $x^{(0)}, \dots, x^{(H)}$ from x_k with u_b satisfies (6) for ϑ . Two cases. (a) The module returns a candidate certified on $\Theta'_k \ni \vartheta$ (Lemma 2); then (6) holds for ϑ from $x_{k+1} = x^{(0)}$, which is $(x_{k+1}, u_k) \in \mathcal{S}(\{\vartheta\})$. (b) No candidate is certified on Θ'_k and the module returns u_b ; then $x_{k+1} = x^{(0)}$ of the rollout verified at step k , and the backup rollout from x_{k+1} is its tail $x^{(1)}, \dots, x^{(H)}$ followed by one more backup step $x^{(H+1)}$. The tail satisfies $c \geq m$ by (6), $x^{(H)} \in \mathcal{C}$ with $c(x^{(H)}) \geq m$, hence by Assumption 1 $c(x^{(H+1)}; \vartheta) \geq 0$ and the closed loop from $x^{(H)}$ stays in $\mathcal{C}$, so the new rollout satisfies (6) for ϑ up to the margin at the last step, which is what the induction needs (the margin m is required only where the sampled verification is invoked, the last step being covered by the assumption). Case (b) also shows that the returned command is certified on $\{\vartheta\}$. $\square$

Lemma 4 (Extension to the box). *Let ℓ_ϑ be a Lipschitz constant of $\vartheta \mapsto \min_j c(x^{(j)}; \vartheta)$ over the rollouts of Definition 1 and $d_N = \max_{\vartheta \in \Theta} \min_i \|\vartheta - \vartheta_i\|$ the covering radius of Θ_N . If a candidate is certified on Θ_N with margin $m \geq \ell_\vartheta d_N$, it is certified with margin 0 on the whole box Θ .*

Proof. For $\vartheta \in \Theta$ pick ϑ_i with $\|\vartheta - \vartheta_i\| \leq d_N$; then $\min_j c(x^{(j)}; \vartheta) \geq \min_j c(x^{(j)}; \vartheta_i) - \ell_\vartheta d_N \geq m - \ell_\vartheta d_N \geq 0$. $\square$

4.4 Policy and its Lipschitz constant

The policy is $u_k^{\text{NN}} = \pi_\phi(w_k)$ with the window $w_k = (x_{k-L+1}, u_{k-L}, \dots, x_k, u_{k-1}, \rho_t)$. The bounded head is

$$\theta_{\text{sp}} = \bar{\theta}_{\text{sp}} \tanh(y_1), \quad T_i^c = \text{sat}_{[0, \bar{T}_i]}(T_{i,k-1}^c + \bar{\delta}_i \tanh(y_{i+1})), \quad y = \text{MLP}(z_{\text{ctx}}, z_L, \rho_t, u_{k-1}), \quad (9)$$

so $u_k^{\text{NN}} \in \mathcal{U}_k$ by construction.

Lemma 5 (Lipschitz bound). *Let each attention block have H heads with weight matrices W_Q^h, W_K^h, W_V^h, W_O , LayerNorm with gain g_{ln} and input norm bounded by r on the operating windows, and each feed-forward block weights W_1, W_2 with 1-Lipschitz activation. Then π_ϕ is Lipschitz on the operating windows with*

$$\ell_\pi \leq \max\{\bar{\theta}_{\text{sp}}, \bar{\delta}_c, \bar{\delta}_r\} \|W_{\text{mlp}}\|_{\text{prod}} \prod_{n=1}^N \left(1 + \|W_O\|_2 \sum_h \|W_V^h\|_2 (1 + 2r^2 \|W_Q^h\|_2 \|W_K^h\|_2 / \sqrt{d_h})\right) \left(1 + \|W_2\|_2 \|W_1\|_2\right) \|W_{\text{emb}}\|_2 \quad (10)$$

with $\|\cdot\|_{\text{prod}}$ the product of the spectral norms of the MLP layers and the LayerNorm gains absorbed into r .

Proof. Each residual block is $1 + \ell$ Lipschitz when its branch is ℓ -Lipschitz; the feed-forward branch has $\ell \leq \|W_2\|_2 \|W_1\|_2$; a self-attention branch restricted to inputs of norm at most r is Lipschitz with constant $\|W_O\|_2 \sum_h \|W_V^h\|_2 (1 + 2r^2 \|W_Q^h\|_2 \|W_K^h\|_2 / \sqrt{d_h})$, the second term bounding the sensitivity of the softmax weights (its Jacobian has spectral norm at most $1/2$, multiplied by the query–key scores’ Lipschitz constant $2r \|W_Q^h\|_2 \|W_K^h\|_2 / \sqrt{d_h}$). The embedding is linear, the MLP head is a product of spectral norms, and $\tanh$ is 1-Lipschitz scaled by the output ranges; sat is 1-Lipschitz. Composition multiplies the constants. $\square$

4.5 Main result

Let $F(x, u; \vartheta)$ be the one-step map of (1), $x^t(\rho_t)$ the trim at the target airspeed and $W(x) = \|x - x^t(\rho_t)\|_P^2$, $P \succ 0$. The closed loop is $x_{k+1} = F(x_k, u_k; \vartheta) + w_k$, $\|w_k\| \leq \bar{w}$, with u_k from (7) applied to $u_k^{\text{NN}} = \pi_\phi(w_k)$.

Table 1: Parameter box Θ identified from the flight log (nominal value and interval).

parameter	nominal	interval	parameter	nominal	interval
C_{L0}	1.141	± 0.144	κ [1/s]	2.0	[0.67, 5.0]
$C_{L\alpha}$ [1/rad]	3.12	$\pm 25\%$	g_θ	1.0	[0.9, 1.0]
C_{D0}	0.133	± 0.044	λ_c [1/s]	2.0	[1.33, 4.0]
k	0.060	$\pm 25\%$	λ_r [1/s]	6.67	[4.44, 13.3]
k_c [N]	30.3	$\beta_c \in [0.78, 1.22]$	k_r [N]	118.7	$\beta_r \in [0.85, 1.15]$

Definition 2 (Verified decrease). *The trained closed loop satisfies the decrease condition with constants (c, β, d) if on a grid $\mathcal{G}_W$ of spacing d covering $\{x : W(x) \geq \beta\} \cap \mathcal{X}$ and for every $\vartheta \in \Theta_N$, $W(F(x_i, u(x_i); \vartheta)) - W(x_i) \leq -cW(x_i) - \ell_W d$, where $u(\cdot)$ is the filtered network command and ℓ_W a Lipschitz constant of $x \mapsto W(F(x, u(x); \vartheta)) - W(x)$, computable from ℓ_x, ℓ_u, ℓ_π and the Lipschitz constant of the filter map $u^{\text{NN}} \mapsto u$ on the segment (7).*

Theorem 1 (Safe generalised hand-off). *Let the true plant $\vartheta \in \Theta_N$, Assumption 1 hold, the measurement errors satisfy $\varepsilon, \bar{w} = 0$ and $(x_0, u_{-1}) \in \mathcal{S}(\Theta_N)$. Then for every command sequence of the network:*

- (i) (safety) $c(x_k; \vartheta) \geq 0$ for all $k \geq 0$: the lift fraction stays above $1 - K_L$, $\alpha_k \leq \bar{\alpha}$, θ_k and γ_k stay in their boxes, the sink rate above $-\dot{\gamma}_{\text{lim}}$, and $u_k \in \mathcal{U}_k$; the module never runs out of certified commands;
- (ii) (practical convergence) if in addition Definition 2 holds with (c, β, d) , then $W(x_k) \leq \max\{(1-c)^k W(x_0), \beta\}$; with disturbances $\bar{w} > 0$, $\limsup_k W(x_k) \leq \max\{\beta, (2\|P\|_2 \bar{w} \bar{r} + \|P\|_2 \bar{w}^2)/c\}$, $\bar{r} = \max_{x \in \mathcal{X}} \|x - x^t\|$;
- (iii) (intervention) $u_k = u_k^{\text{NN}}$ whenever the network command is certified on Θ'_k ; the intervention rate of the trained policy is a measured quantity;
- (iv) (whole box) if the certificates hold with the margin of Lemma 4, (i) holds for every $\vartheta \in \Theta$.

Proof. (i) By Lemma 3, u_k is certified on $\{\vartheta\}$ at every step, so $c(x_{k+1}; \vartheta) = c(x^{(0)}; \vartheta) \geq m > 0$ by (6); $u_k \in \mathcal{U}_k$ because both u_k^{NN} (bounded head (9)) and u_b (saturation in (5)) lie in the convex set $\mathcal{U}_k$ and (7) is a convex combination. The module always has u_b available by Lemma 3. (ii) On the cell of $x_i \in \mathcal{G}_W$, $W(x_{k+1}) - W(x_k) \leq -cW(x_k) - \ell_W d + \ell_W \|x_k - x_i\| \leq -cW(x_k)$ whenever $W(x_k) \geq \beta$, hence $W(x_{k+1}) \leq (1-c)W(x_k)$ until the level β is reached, after which $W(x_{k+1}) \leq (1-c)\beta \leq \beta$. With disturbances, $W(F+w) - W(F) = 2(F - x^t)^\top Pw + w^\top Pw \leq 2\|P\|_2 \bar{r} \bar{w} + \|P\|_2 \bar{w}^2 =: \Delta_w$, so $W(x_{k+1}) \leq (1-c)W(x_k) + \Delta_w$ and $\limsup_k W(x_k) \leq \Delta_w/c$ when this exceeds β . (iii) If u_k^{NN} is certified, $\lambda^* = 1$ in (7). (iv) Lemma 4 applied to every certificate used in Lemma 3. $\square$

Proposition 1 (Identifiability from the window). *On a window of L samples with α not constant and V not constant, the aero parameters $(C_{L0}, C_{L\alpha}, C_{D0}, k)$ enter the finite-difference residuals of (1) linearly through the regressors $\bar{q}S(1, \alpha)$ and $\bar{q}S(1, C_L^2)$; the least-squares estimate from the window is unique iff the regressor matrix has full column rank, and its error is bounded by $\sigma\sqrt{L}/\sigma_{\min}(\Phi_L)$ for measurement noise of variance σ^2 . The attention block with the context token can represent this estimator to any accuracy on compact windows (universal approximation of continuous maps), which is the target of the auxiliary head.*

Proof. Rearranging (1), $mV\dot{\gamma} - T_c \sin \alpha - T_r \cos \alpha + mg \cos \gamma = \bar{q}S(C_{L0} + C_{L\alpha}\alpha)$ and $-m\dot{V} + T_c \cos \alpha - T_r \sin \alpha - mg \sin \gamma = \bar{q}S(C_{D0} + kC_L^2)$, linear in the parameters with the stated regressors; stacking L samples gives $\Phi_L \vartheta_a = y_L + e$, whose least-squares solution is unique iff $\text{rank } \Phi_L = 4$, i.e. α and C_L^2 vary over the window, and $\|\hat{\vartheta}_a - \vartheta_a\| \leq \|\Phi_L^\dagger\| \|e\| \leq \sigma\sqrt{L}/\sigma_{\min}(\Phi_L)$. The representability claim is the universal approximation property of the transformer block with a readout token on compact sets. $\square$

Algorithm 1 Physics-informed training of HOT

- 1: build the differentiable model F of (1) (RK4, Δ), the box Θ , the rows (??)
 - 2: **for** epoch = 1, . . . , E **do** (horizon N from the curriculum)
 - 3: sample a batch of plants $\vartheta \sim \mathcal{U}(\Theta)$, gust signals, noise levels, initial trims ρ_0 and targets ρ_t
 - 4: roll out $x_{j+1} = F(x_j, \tilde{u}_j; \vartheta) + w_j$, $\tilde{u}_j = \pi_\phi(w_j)$ with the soft barrier of the LSM rows, $j = 0, \dots, N-1$
 - 5: $\mathcal{J} = \sum_j [\text{time} + \text{energy} + \text{barrier} + \text{plant head} + \text{smoothness}]$ (Fig. 3)
 - 6: $\phi \leftarrow \phi - \eta_{\text{lr}} \nabla_\phi \mathcal{J}$ by backpropagation through time
 - 7: **end for**
 - 8: verify Definition 2 on the grid; report the constants (c, β, d) and the Lipschitz bound (10)
-

Algorithm 2 Online execution (one sample, 50 ms)

- 1: receive x_k from the autopilot; append (x_k, u_{k-1}) to the window w_k
 - 2: $u_k^{\text{NN}} \leftarrow \pi_\phi(w_k)$ by (9)
 - 3: compute the enclosure of Definition ?? (Picard iteration (??), at most six passes) and the rows $A(x_k), b(x_k)$ of (??)
 - 4: **if** $A(x_k)u_k^{\text{NN}} \leq b(x_k)$ **then** $u_k \leftarrow u_k^{\text{NN}}$ **else** $u_k \leftarrow$ solution of the QP (??)
 - 5: send θ_{sp} and the rotor collective as an attitude target and the pusher command as an actuator command
-

5 Algorithms

6 Experiment plan

Table 2: Controllers compared (all on the same simulator, same metrics).

label	controller	learned	safety guarantee
THR	autopilot threshold rule	no	no
SCH	fixed lift-sharing schedule	no	no
GS-PID	gain-scheduled PID on airspeed and pitch	no	no
MPC	online nonlinear MPC on the nominal model, 4 s horizon	no	nominal only
MLP	feed-forward policy on the current state	yes	no
LSTM	recurrent policy on the window	yes	no
HOT–LSM	the transformer without the safety module	yes	no
PPO, SAC	reinforcement learning on the same simulator with the barrier reward	yes	no
HOT	this paper	yes	Theorem 1

7 Where the author’s published works are cited

Dong2020 (adaptive VTOL control, related work 2.1); Dony2020acc, Dony2021jae, Dony2021thesis (robust tracking and formation control under parametric uncertainty, related work 2.2, contrast: no learning, no state constraints); Dony2025rs, Dony2025ieom, Dony2026cdc (safe learning with control barrier functions and model-free Q-learning, related work 2.2, contrast: filters a nominal command instead of learning a policy with a proof over a plant family); Dony2026data (the flight data set, Section 7, cited as [dataset]). The TCST, CEP and SciTech manuscripts are not mentioned.

Table 3: Test cells and metrics.

cell	content
nominal	identified plant, front ($6 \rightarrow 15$) and back ($15 \rightarrow 6$ m/s)
cross-plant	100 plants drawn from Θ , never seen in training; per-plant metrics, Wilcoxon signed-rank against HOT, Friedman ranks
out-of-box	50 plants with parameters 25 % beyond the box faces (extrapolation)
gusts, noise	vertical gusts of 1–4 m/s, airspeed noise of 0.2–1 m/s, 20 seeds each
replay	the states recorded in flight as initial conditions
PX4 SIL	the real firmware in the loop at real time, both transitions
metrics	time to target, energy, minimum lift fraction, maximum α , violation ticks, LSM intervention rate, success rate, update time
statistics	mean $\pm$ std over plants and seeds; Wilcoxon $+/-$ counts at $p < 0.05$; Friedman mean ranks; training convergence curves
ablations	–physics features, –PIA (MLP on the window), –auxiliary head, –BPH (clipped outputs), –LSM; window $L \in \{10, 20, 40, 80\}$, width $\{32, 64, 128, 256\}$, depth $\{1, 2, 4\}$
interpretability	attention maps over the window; context-token estimates against the sampled parameters (Proposition 1)

8 Journal compliance

Editable LaTeX (elsarticle, double column allowed), figures as separate PDF files, abstract ≤ 250 words, 1–7 keywords, highlights file (3–5 bullets, ≤ 85 characters), numbered sections, references numbered in order of appearance with the dataset tagged [dataset], CRediT statement, competing-interest declaration (.docx from the Elsevier tool), funding sentence, generative-AI declaration before the references, data-availability statement with the dataset DOI, vitae with photograph. No page limit is stated for regular articles; the target is 25–30 manuscript pages in the double-column template.

9 Approval points

1. the name HOT and the four module names (SWE, PIA, BPH, LSM);
2. the safety design: rows (??) with $K_L = 0.35$ and $\eta = 0.5$, and the safe set of Fig. 3 here;
3. the training route: physics-informed backpropagation through the identified model, no expert controller;
4. the theorem set: Lemmas 1–6, Theorem 1(i)–(iii), Proposition 1;
5. the baseline list (nine controllers) and the test cells;
6. the figure list and the style (module diagrams, ablation bars, sweeps, attention maps, photographs).